

A Scientific Visualization Platform for Fire Dynamics Simulator Ensembles

Internship Progress Report

1. Project Overview

The Fire Dynamics Simulator (FDS) is a software used to simulate how fire, heat, and smoke move inside buildings. In real studies, researchers usually run many simulations with different conditions, such as changing the fire size, ventilation, or door openings. The main challenge is not only running these simulations, but also comparing and understanding the results.

This project focuses on developing an application that helps researchers visualize and analyze FDS simulations. It allows them to follow the evolution of physical quantities, compare different scenarios, and extract useful information from the simulation data.

2. Initial State of the Project

At the beginning of the internship, the application already existed but had limited capabilities. It could display the temperature evolution of a single simulation using a basic animated heat map. However, it supported only one quantity and one scenario at a time, with no measurement tools, comparison features, or advanced analysis capabilities.

The objective was to transform it into a modern scientific visualization tool while keeping the existing structure and improving it step by step instead of rebuilding the whole application.

3. Work Performed

Data loading. Improved the way simulation data is loaded by introducing optimized reading methods and caching. This made switching between scenarios faster and provided a better foundation for interactive visualization and analysis.

Playback. Improved the animation system to make it more useful for studying fire evolution. Users can now move freely through time using a timeline, go frame by frame, change playback speed, and compare several simulations at the same moment.

Scientific visualization. Added support for multiple physical quantities and new visual analysis tools. The application now includes side-by-side scenario comparison, difference views, ensemble views, interactive probes, isotherm and contour visualization, and extraction of information from volumetric smoke data.

Analysis. Added several tools to study groups of simulations instead of only individual cases. These include a dataset explorer, statistical summaries, time-series plots at selected locations, principal component analysis for identifying similar fire behaviors, forecasting models, and fire-related indicators such as energy release, smoke-layer height, and safety conditions. The application can also generate reports and support automated execution through a command-line interface.

User interface. Redesigned the interface to make it easier to use during a research workflow. The application is now organized around the main tasks: viewing a simulation, comparing cases, exploring datasets, performing analysis, and exporting results. Improvements include clearer

controls, better visualization of the physical setup, improved typography, and support for both light and dark modes.

Software architecture. Improved the internal organization of the application by separating data processing, application logic, and visualization components. A common system was created for handling different quantities and views, making it easier to add future features while keeping the software maintainable.

4. Final Application

The final result is no longer only a visualization tool. It has become a complete platform for exploring and analyzing FDS simulation studies.

Researchers can load simulation cases, observe how fire conditions change over time, compare multiple scenarios, identify important differences, explore datasets, analyze patterns, and generate reports. The application is also designed to support future FDS outputs and can be used without always relying on the graphical interface.

5. Future Work

The platform provides a strong foundation for future improvements, especially in the area of artificial intelligence. Some data-driven methods are already included, such as forecasting and grouping simulations based on their behavior.

Future developments could include more AI-based assistance, such as automatically identifying fire behaviors, helping researchers interpret simulation results, or exploring large numbers of simulation parameters more efficiently. These directions will be studied and developed in future work.

6. Conclusion

This internship transformed a simple FDS visualization application into a more complete scientific analysis platform. The project introduced new visualization, comparison, measurement, and analysis capabilities while keeping the original system and improving it progressively.

The final application provides researchers with a more efficient way to study fire simulations and creates a solid foundation for future developments, including the integration of more advanced artificial intelligence techniques.